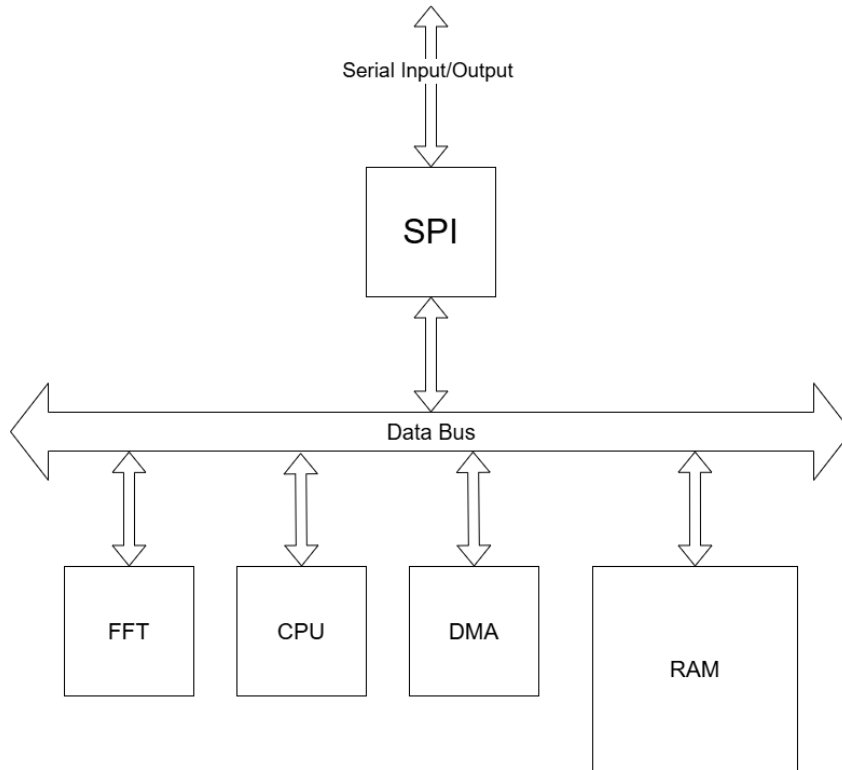


Audio Spectrum Analyzer SoC Lab

1. Architecture



2. Hardware / Software Partitioning

Task	HW or SW	Block	Reason
Receive serial samples	HW	SPI	SPI hardware handles the timing of the serial protocol so the CPU does not have to.
Move samples to RAM	HW	DMA	DMA moves the data automatically, so the CPU is free to do other things.
Store samples	HW	RAM	This is just memory storage.
Windowing	SW	CPU	Simple multiply operation, easy to keep flexible in software.
Compute FFT	HW	FFT block	The FFT is heavy math, so a dedicated hardware block does it much faster than software.
Compute magnitude spectrum	SW	CPU	Simple calculation per value, not worth building special hardware for it.
Find peak bin	SW	CPU	Just a simple search loop, easy for the CPU.
Convert bin to frequency	SW	CPU	Simple math formula, best done in software.
Send result back	HW	SPI	SPI hardware also handles sending the answer back to the host.

2. System Application Pseudocode

```
BEGIN

// 1. Receive input
WAIT for SPI interrupt (data ready)
DMA_transfer(SPI_RX_buffer -> RAM_sample_buffer, N samples)

// 2. windowing
FOR i = 0 to N-1:
    sample_buffer[i] = sample_buffer[i] * window_coeff[i]

// 3. Compute FFT (hardware call)
FFT_start(sample_buffer, N)
WAIT for FFT_done interrupt
fft_result = FFT_read_output()

// 4. Compute magnitude spectrum
FOR i = 0 to N/2 - 1:
    magnitude[i] = sqrt(real[i]^2 + imag[i]^2)

// 5. Find peak bin
peak_index = 0
peak_value = magnitude[0]
FOR i = 1 to N/2 - 1:
    IF magnitude[i] > peak_value:
        peak_value = magnitude[i]
        peak_index = i

// 6. Convert bin to frequency
peak_freq = peak_index * (Fs / N)

// 7. Send result back to host
SPI_send(peak_freq)

END
```